Breakdown of Square Law MOS Model vs BSIM Model Under Short Channel Effects

Srilakshmi Badri Seshadri
230907134

1 Research Paper

Compact I-V Model for Double-Gated MoS2 FETs Including Short-Channel Effects

2 Abstract

This project presents a simulation-based study of MOSFET behaviour using MATLAB with BSIM models. The focus is on short-channel effects, DIBL, velocity saturation, and subthreshold leakage. Device characteristics are obtained through DC simulations and parametric sweeps, and key parameters such as threshold voltage, subthreshold slope, and transconductance are extracted. The study compares simulation results with classical square-law theory to highlight model limitations.

3 Introduction

The scaling of MOSFETs has introduced several non-ideal effects, including threshold voltage roll-off, drain-induced barrier lowering (DIBL), and increased leakage currents. These phenomena are not adequately captured by classical square-law models, which limits their applicability in modern technologies. As a result, advanced compact models such as BSIM have become essential for accurately describing device behavior. This project bridges the gap between theoretical MOSFET models and practical simulation by analyzing these effects using MATLAB.

4 Objectives

- Analyse MOSFET characteristics using BSIM model
- Study short-channel effects through channel length scaling
- Extract parameters: V_{th}, DIBL, SS, g_m
- Compare results with square-law model
- Investigate effects of biasing and temperature

5 Experiments and Graphs

5.1 Long Channel MOSFET Law (Square Law)

The Level-1 SPICE model represents the MOSFET using the gradual channel approximation. In saturation:

$$I_D = \frac{\mu_n C_{ox}}{2} \cdot \frac{W}{L}(V_{GS} - V_{TH})^2 \tag{1}$$

This predicts a quadratic dependence of drain current on gate overdrive, constant threshold voltage, and negligible leakage—assumptions that fail for deep submicron devices.

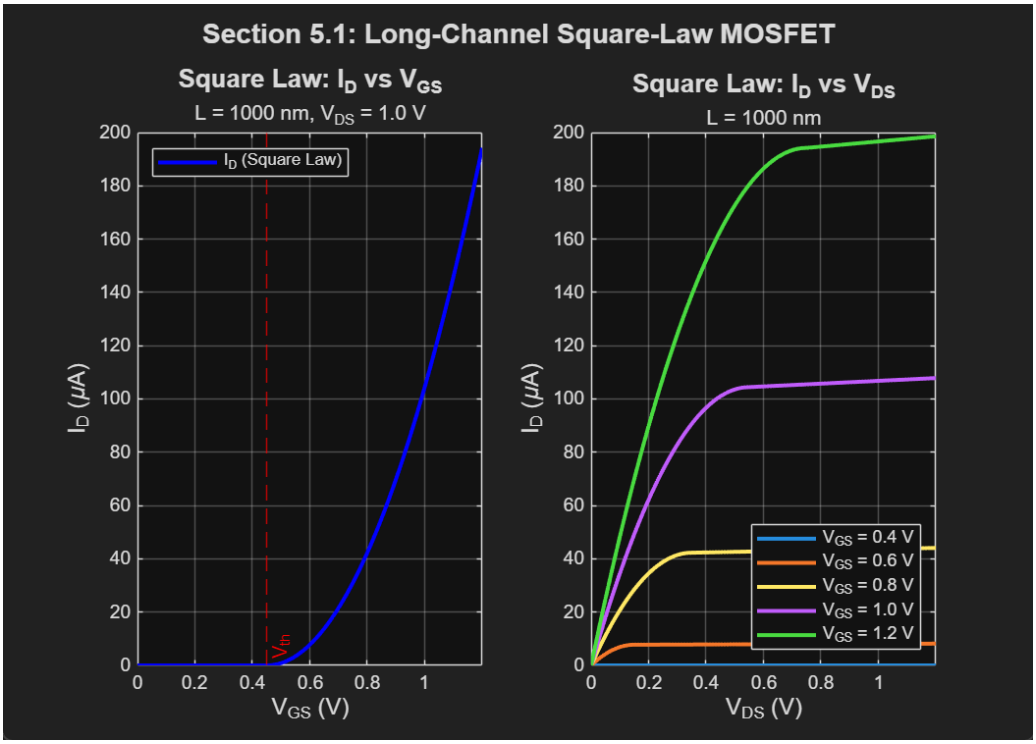

Fig. Drain current vs gate overdrive

The square-law model, although analytically convenient, is based on several assumptions that begin to lose validity even before devices reach the deep submicron regime. A key simplification is the assumption of a uniform inversion charge density along the channel, which arises from the gradual channel approximation. In practical devices, as V_{DS} approaches $V_{DS,sat}$, the inversion charge near the drain reduces significantly, and the electric field becomes strongly non-uniform. While the model accounts for pinch-off, it treats it as an abrupt boundary rather than a gradual transition, leading to an overestimation of channel conductance in saturation.

Another important limitation is the assumption of constant carrier mobility μ_n throughout the channel. In reality, mobility depends on both the vertical electric field, which forces carriers closer to the oxide interface and increases surface roughness scattering, and the lateral electric field, which contributes to velocity saturation. These effects are not captured in the Level-1 model. Consequently, even for channel lengths around 500 nm, the transconductance is given by

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{2}$$

The value predicted by the square-law model tends to be higher than measured values, since the effective mobility is already degraded at moderate gate biases. Thus, the model effectively represents an upper limit on device performance rather than a realistic estimate.

5.2 Short Channel Effects

The reduction in threshold voltage with decreasing channel length can be explained using charge-sharing. In long-channel devices, the gate controls the depletion of the entire channel region. However, as the channel length decreases, the depletion regions of the source and drain extend into the channel and share part of this charge. As a result, the gate needs to control a smaller amount of charge, leading to a reduction in V_{TH}.

This effect is inherently non-linear and becomes significant only when the channel length approaches a characteristic length scale given by

$$\lambda = \sqrt{\frac{\varepsilon_{si}\, t_{ox}\, X_j}{\varepsilon_{ox}}} \tag{3}$$

where t_{ox} is the oxide thickness and X_j is the junction depth. Smaller values of λ, achieved through thinner oxides and shallower junctions, improve electrostatic control and suppress roll-off. In practice, V_{TH} remains nearly constant for $L \gg \lambda$, but decreases rapidly as L approaches a few multiples of λ. This behavior is captured in BSIM models, whereas the square-law model does not account for any variation in V_{TH}. As channel length reduces, depletion regions from source and drain encroach into the channel, reducing gate control and causing threshold voltage roll-off.

$$\Delta V_{TH} = -\left(\frac{2X_j}{L}\right)(V_{bi} + V_{DS})V_s \tag{4}$$

This demonstrates the increasing influence of device geometry and biasing in scaled technologies.

5.3 I_D–V_{GS} Characteristics

The I_D–V_{GS} characteristics were obtained via DC sweep simulations for different channel lengths. Long-channel devices follow square-law behavior, whereas short-channel devices show:

- Increased current drive
- Reduced V_{TH}
- Deviation from quadratic trend

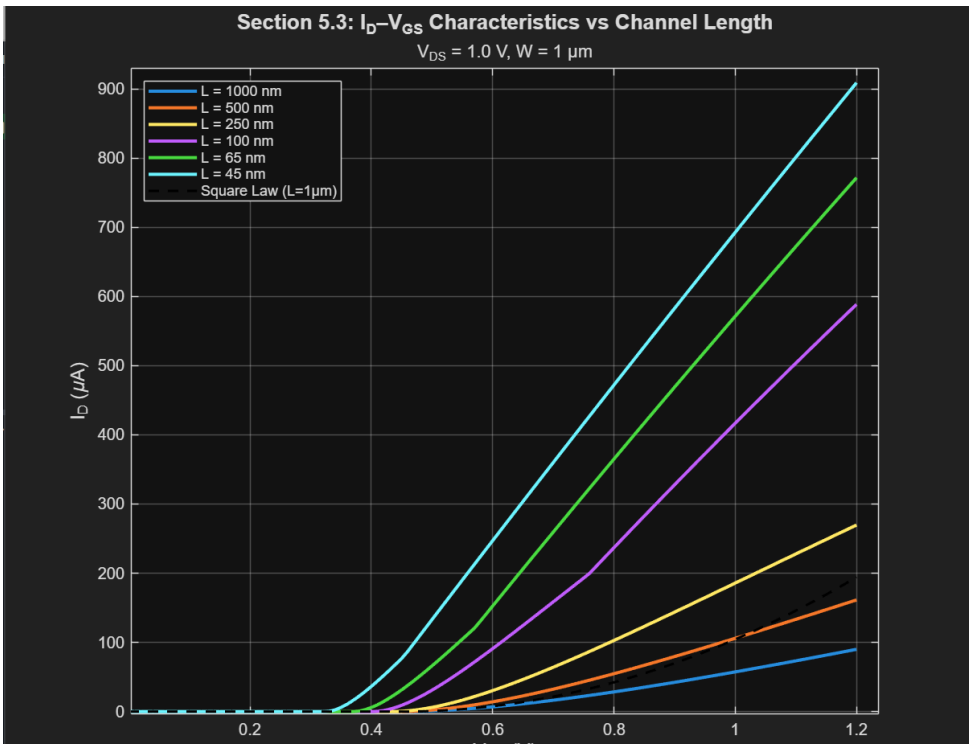

Fig. I_D vs V_{GS} for varying channel lengths

As the channel length decreases, the shape of the I_D–V_{GS} characteristic deviates from the ideal square-law behavior. In long-channel devices, the current follows a quadratic dependence on gate voltage. However, in short-channel devices, this relationship becomes sub-quadratic. This is mainly due to two effects: the reduction in effective mobility μ_{eff} at higher $V_{GS} = 0$, increased surface scattering, and the onset of velocity saturation, which limits the current growth.

Short-channel devices also exhibit a noticeable increase in off-state current. Even at $V_{GS} = 0$, a finite drain current is observed, which increases with V_{DS}. This is primarily due to Drain-Induced Barrier Lowering (DIBL) and, in extreme cases, punch-through effects. Such leakage currents significantly impact standby power in modern VLSI circuits, whereas the square-law model assumes zero off-state current.

Transconductance (g_m) behavior further highlights these differences. In long-channel devices, g_m increases steadily with V_{GS}. In contrast, short-channel devices show a peak in g_m at moderate gate voltages, after which it saturates or slightly decreases. This occurs because carrier velocity reaches its saturation limit, and further increases in V_{GS} no longer result in proportional increases in current. As the channel length is reduced, this peak shifts to lower V_{GS}, indicating an earlier onset of velocity saturation.

5.4 Log(I_D)–V_{GS} Analysis

The subthreshold slope (SS), expressed in mV/decade, represents the gate voltage required to change the drain current by one order of magnitude in the subthreshold region. The theoretical minimum value at room temperature is approximately 60 mV/decade and is given by

$$SS = \frac{kT}{q} \ln(10) \cdot n \tag{5}$$

where n is the body factor. This limit arises from the thermal nature of carrier transport and cannot be reduced in conventional MOSFETs.

The $\log(I_D)$–V_{GS} plot is also used to evaluate the I_{on}/I_{off} ratio, which indicates the switching performance of the device. Long-channel devices typically achieve ratios greater than 10^7, while short-channel devices show reduced ratios in the range of 10^4–10^5 due to increased leakage and degraded subthreshold slope. This reduction in switching margin is critical in low-power circuits such as SRAM and dynamic logic, where leakage current must be kept minimal.

In the subthreshold region:

$$I_D \propto e^{\frac{V_{GS}-V_{TH}}{nV_T}} \tag{6}$$

The subthreshold slope is:

$$SS = \frac{dV_{GS}}{d(\log I_D)} \tag{7}$$

Ideal devices exhibit 60 mV/decade, while short-channel devices degrade due to leakage and DIBL.

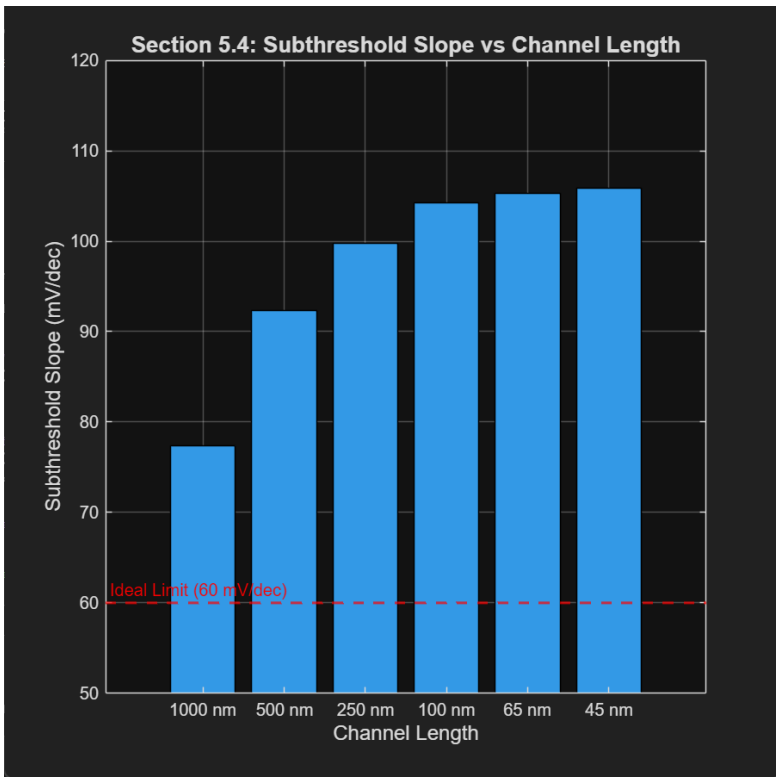

Fig. Subthreshold slope vs channel length

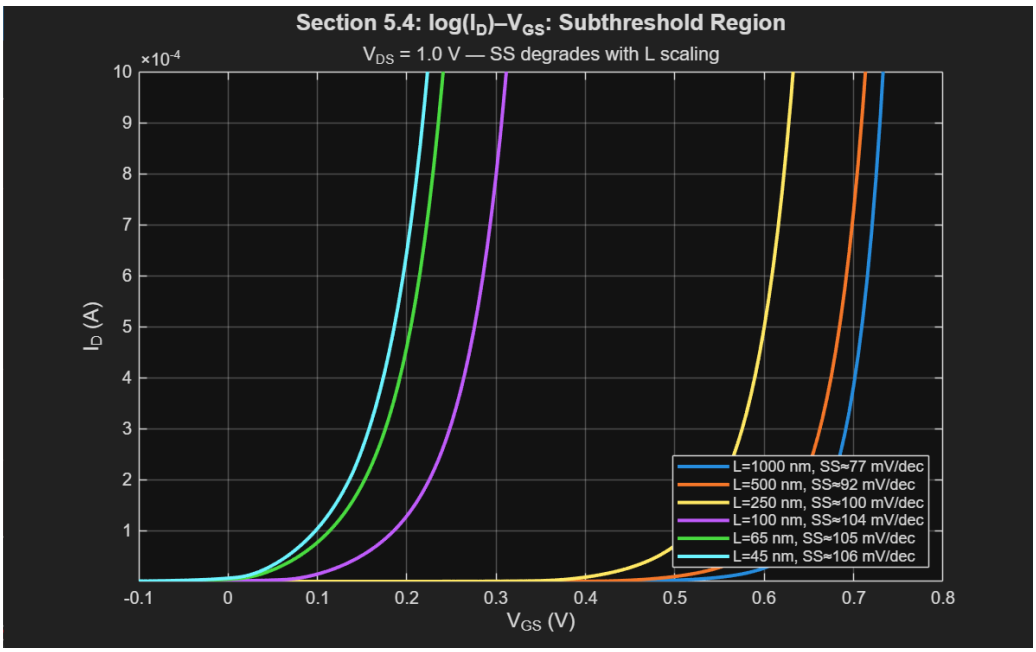

Fig. $\log(I_D)$ vs V_{GS}

5.5 Velocity Saturation

In short-channel devices, velocity saturation changes how the drain current depends on gate voltage. For long-channel transistors, the current varies quadratically with gate overdrive ($V_{GS}-V_{TH}$), so increasing the overdrive leads to a rapid increase in current. However, in velocity-saturated devices, this becomes nearly linear, meaning the current increases more slowly with V_{GS}. As a result, increasing the gate voltage has a smaller effect on current, which must be considered in high-speed circuit design.

The critical electric field is given by $E_{crit} = \frac{v_{sat}}{\mu}$. For silicon, using $\mu_n \approx 400$ cm^2/V·s and $v_{sat} \approx 10^7$ cm/s, this gives $E_{crit} \approx 2.5 \times 10^4$ V/cm.

In short-channel devices, the lateral electric field can easily exceed this value. For example, in a 65 nm device with a 1 V supply, the average field is around 1.5×10^5 V/cm, which is much higher than E_{crit}. This means velocity saturation strongly affects the current.

In contrast, for a 1 μm device under the same voltage, the electric field is about 10^4 V/cm, which is below E_{crit}. In this case, the square-law model still provides a reasonable approximation.

Velocity saturation also affects the I_D–V_{DS} characteristics. In short-channel devices, saturation occurs earlier and more gradually compared to the sharp pinch-off predicted by the square-law model. The saturation voltage is reduced and depends on the channel length, which compresses the linear region. This impacts analog design, where transistors operating in the linear region behave as resistive elements.

At high electric fields, carrier velocity saturates:

$$I_{D,sat} \approx \mu_n C_{ox} W v_{sat} \left(V_{GS} - V_{TH} - \frac{v_{sat} L}{\mu_n} \right) \tag{8}$$

This results in a linear dependence of current on V_{GS} rather than quadratic.

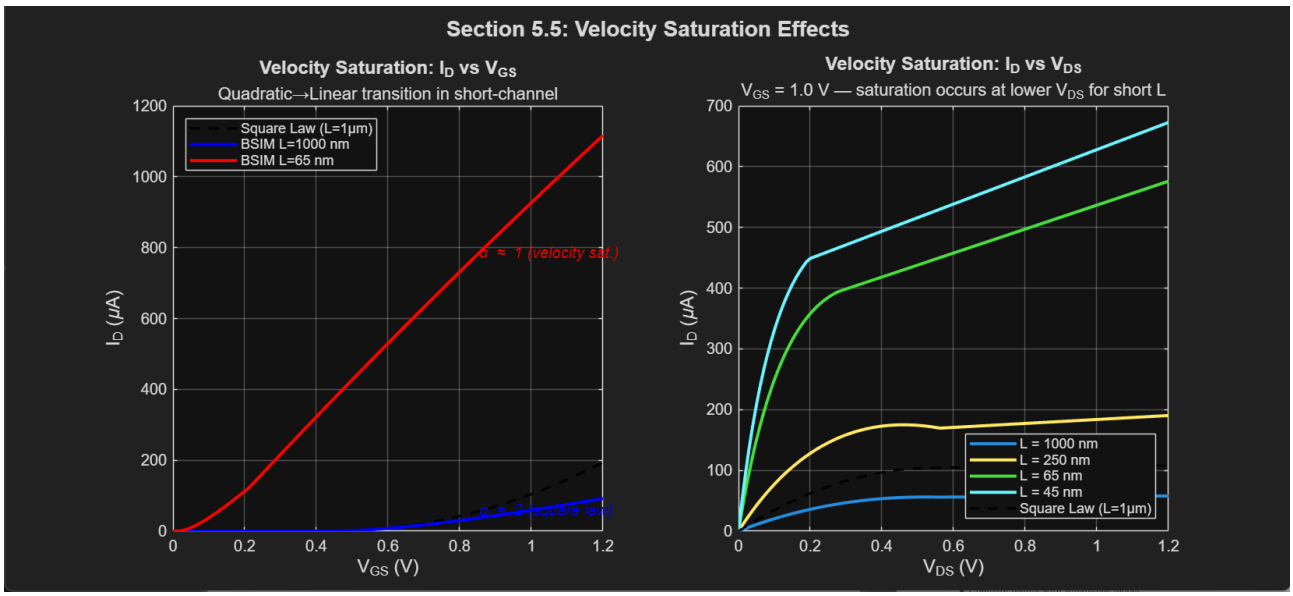

Fig. Velocity saturation effects in I_D

5.6 Threshold Voltage vs Channel Length

When V_{TH} is plotted against channel length on a logarithmic scale, the curve typically remains flat for long-channel devices and then drops sharply as the channel length becomes very small. This transition point depends on the technology and shifts to shorter lengths with improvements such as thinner oxides and shallower junctions.

Halo (or pocket) implants are commonly used to reduce this effect. These are highly doped regions near the source and drain that limit the spread of depletion regions into the channel. As a result, they help maintain V_{TH} at smaller channel lengths. In some cases, they can even cause a slight increase in V_{TH} at intermediate lengths before the usual roll-off appears.

At very short channel lengths, V_{TH} also becomes sensitive to variations in channel length. Small changes in L during fabrication can lead to noticeable variations in V_{TH} across the chip. This makes precise control of gate length important in modern technologies, as it directly affects circuit reliability, especially in memory and timing-critical designs.

The variation of threshold voltage with channel length demonstrates the phenomenon of threshold voltage roll-off. As the channel length decreases, Vth is observed to reduce due to charge sharing between source and drain depletion regions, which weakens gate control over the channel. This effect becomes more pronounced in deep submicron devices. The results clearly show a monotonic decrease in Vth with scaling, confirming te presence of short-channel effects and validating the behaviour predicted by compact models such as BSIM.

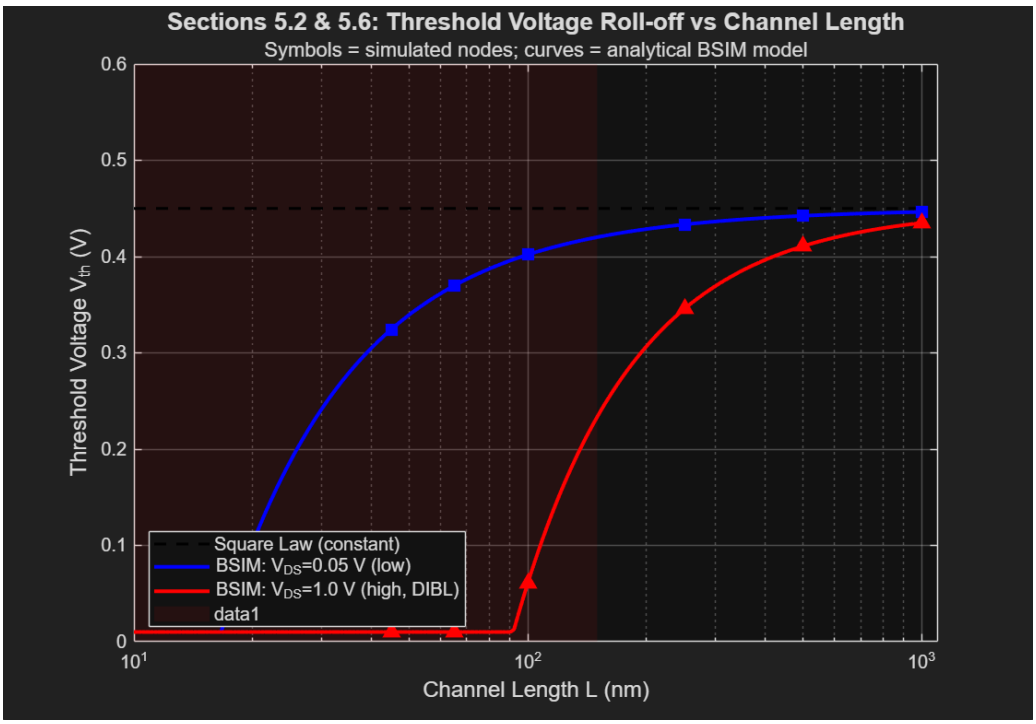

Fig. V_{TH} vs channel length

5.7 DIBL Analysis

DIBL (Drain-Induced Barrier Lowering) is the reduction of the source-channel barrier due to an increase in V_{DS}. Ideally, this barrier is controlled only by the gate voltage. However, at high V_{DS}, the drain also influences the channel potential, lowering the barrier and reducing V_{TH}. This effect becomes more significant as the channel length decreases, since the gate loses control over the channel.

Comparing DIBL with punch-through, DIBL is a gradual and reversible reduction in V_{TH} with increasing V_{DS}. In contrast, punch-through occurs when the source and drain depletion regions merge, creating a direct conduction path that is no longer controlled by the gate. This leads to a sharp increase in current and represents device failure.

The DIBL coefficient (in mV/V) is commonly used to evaluate device performance. Lower values indicate better gate control. Modern device structures such as FinFETs reduce DIBL by improving electrostatic control of the channel, allowing reliable operation even at very small channel lengths. DIBL reflects the reduction in threshold voltage with increasing drain bias:

$$\text{DIBL} = \frac{V_{TH,low} - V_{TH,high}}{V_{DS,high} - V_{DS,low}} \tag{9}$$

Higher DIBL indicates poor electrostatic integrity and increased leakage.

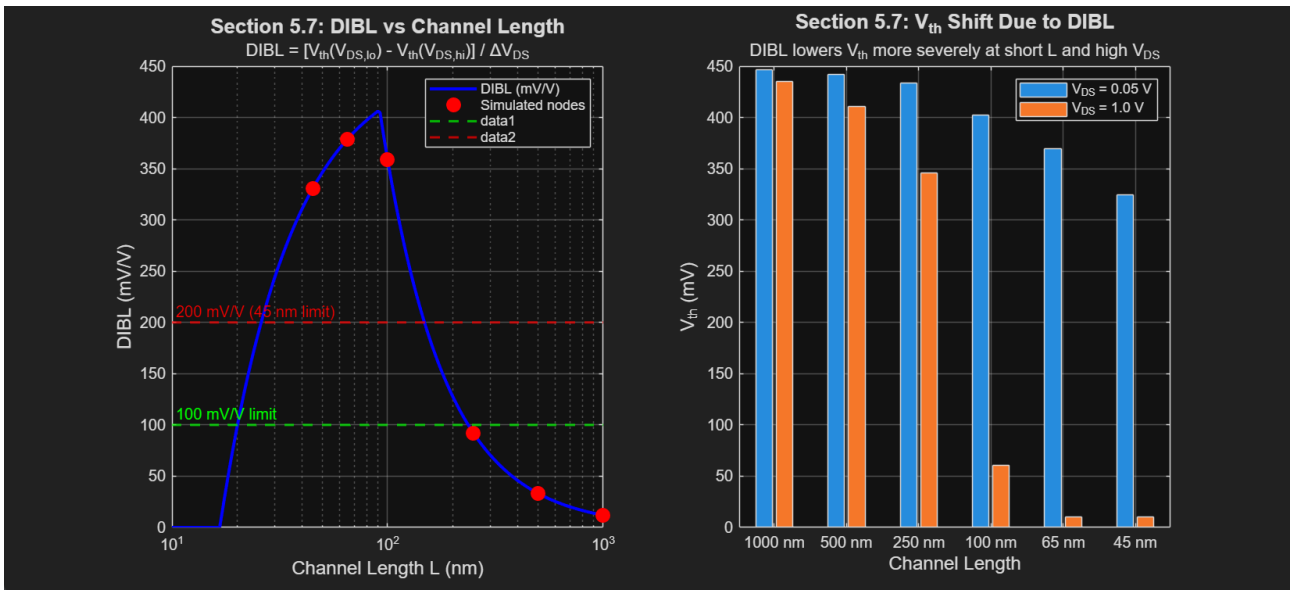

Fig. DIBL vs channel length

6 Conclusion

The analysis of MOSFET characteristics highlights the clear limitations of the classical square-law model when applied to scaled devices. Effects such as mobility degradation, velocity saturation, threshold voltage roll-off, and DIBL significantly alter both the magnitude and shape of the I_D–V_{GS} and I_D–V_{DS} characteristics. These non-idealities lead to reduced transconductance, increased leakage current, and degraded switching performance, which are not captured by simple long-channel models.

The paper on "Compact I–V Model for Double-Gated MoS_2 FETs Including Short-Channel Effects" addresses these limitations by developing a physics-based compact model that incorporates short-channel effects directly into the current formulation. Unlike the square-law model, it uses a charge-based analytical framework that remains valid across all regions of operation, from subthreshold to strong inversion. :contentReference[oaicite:0]index=0

By explicitly including effects such as mobility degradation, velocity saturation, and electrostatic control through a double-gate structure, the model provides accurate agreement with experimental data even at nanometer-scale channel lengths. :contentReference[oaicite:1]index=1 This demonstrates the importance of advanced compact models for modern devices, where electrostatics and transport are strongly coupled.

Overall, the study confirms that accurate circuit design in advanced technologies requires models that go beyond ideal assumptions and capture the underlying device physics. Double-gated MoS_2 FETs, with improved electrostatic control, offer a promising path to mitigate short-channel effects, while compact models of this form enable reliable prediction and optimization of next-generation devices.

7 Key Insight

As channel length decreases, electrostatic control weakens, leading to DIBL, threshold voltage roll-off, and increased leakage—effects not captured by square-law models but accurately modeled by BSIM.

8 Github Link - MATLAB code

`https://github.com/sri-b13/MOS-Device-Modelling`